

ABSTRACT

Data visualization is the discipline of trying to understand data by placing it in a visual context so that patterns, trends and correlations that might not otherwise be detected can be exposed. **Matplotlib** is the most popular python plotting library. It is a low-level library with a MATLAB like interface which offers lots of freedom at the cost of having to write more code. **Pandas DataFrame** is two-dimensional size-mutable, potentially heterogeneous tabular data structure with labelled axes (rows and columns). A DataFrame is a two-dimensional data structure, i.e., data is aligned in a tabular fashion in rows and columns.

In this project, we combine all the above mentioned **DataFrame, Data visualization** in Python to analyse the data which is stored in csv format. The data is then later imported using **read_csv()** function for visualization of the data in the form of different charts. We took the IPL related data to work with our project.

ACKNOWLEDGMENT

We thank Lord Almighty for the blessings he has bestowed on us which resulted in the finalization of our project.

We would like to express our sincere gratitude to this institution, *St. Kuriakose Public School, Mariamala, Kaduthuruthy* & our Manager **Rev. Fr. BINO CHERIYIL** for the worthy support extended to us.

We also thank and express sincere gratitude to our honourable Principal **Ms. RENJI WILSON CHOCKATTU**, for providing us with best facilities and atmosphere for our project development and its realization. We are grateful to the Head of the Department, **Ms. NILNA BABU**, for her constant encouragement and support.

We are highly obliged to our project coordinators and guides, **Mr. GODWIN JOSEPH** and **Ms. NILNA BABU**, for the valuable guidance and help during the course of this project work. We also like to express our gratitude to the lab staffs who helped us to make the project a success.

This acknowledgement will stand incomplete if our parents, friends, seniors & classmates are not thanked for offering their prayers and support to us, which was inevitable for the realization of this work. Last but not the least we are thankful to each other in this project group for being co-operative, patient and hardworking for the successful completion of this project.

GROUP MEMBERS

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DEVANANDA S

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CHAPTER 1 INTRODUCTION

1.1 GENERAL INTRODUCTION

The world is changing rapidly. The era of industrialization has made humans completely dependent on automation. Today computers are common, cheap and are active in almost every field of science and technology. The robotic brains have made advanced development in the field of science, information, communication, technology and commerce. The main function of managing information like collection of data, classification, storing and summarizing are all done with the help of systems.

As a result, a great deal of man-force is considerably reduced. Thus computers have become a part of our daily life. Schools and colleges are entirely dependent on computers for their smooth and easier management. It reduces the burden of people. It helps to manage great amount of data expansively and with such a great accuracy that because of this, book keeping practice that previously prevailed in schools and colleges have become outdated, and all records today are maintained via computers. This helps in promoting greater efficient.

1.2 INTRODUCTION TO PYTHON

Python is a widely used general-purpose, high level programming language. It was created by Guido van Rossum in 1991 and further developed by the Python Software Foundation. It was designed with an emphasis on code readability, and its syntax allows programmers to express their concepts in fewer lines of code.

Python is a programming language that lets you work quickly and integrate systems more efficiently.

1.3 INTRODUCTION TO PANDAS

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. [pandas] is derived from the term "panel data", an econometrics term for data sets that include observations over multiple time periods for the same individuals. Pandas offers many data structures to handle variety of data.

Pandas is a package commonly used to deal with data analysis. It simplifies the loading of data from external sources such as text files and databases, as well as providing ways of analyzing and manipulating data once it is loaded into your computer. The features provided in pandas automate and simplify a lot of the common tasks that would take many lines of code to write in the basic Python language.

1.4 INTRODUCTION TO PANDAS DATAFRAME

A DataFrame in Pandas stores data in a two-dimensional way. It is a two-dimensional labelled array, which is actually an ordered collection of columns, where columns may store different types of data, e.g., numeric or string or floating point or Boolean type etc.

- i. It has two indexes or we can say, two axes – a row index(axis=0) and column index (axis=1)
- ii. Each value is identifiable with the combination of row index and column index.
- iii. The row index is known as index in general and the column index is called the column-name.
- iv. The indexes can be of numbers or letters or strings.
- v. There is no condition of having all data of same type across columns; its columns can have data of different types.
- vi. You can easily change its values, i.e., it is value-mutable.
- vii. You can add or delete rows/columns in a DataFrame. In other words it is size-mutable.

1.5 INTRODUCTION TO DATA VISUALIZATION

Data visualization basically refers to the graphical or visual representation of information and data using visual elements like charts, graphs, maps, etc.

Data visualization in Python can be done via many packages. One example of such a package is matplotlib. The matplotlib is a Python library that provides many interfaces and functionality for 2D- graphics similar to MATLAB's various forms.

The matplotlib library offers many different named collections of methods; PyPlot is one such interface. PyPlot is a collection of methods within matplotlib which allows user to construct 2D plots and graphs easily and interactively. matplotlib is a 2D plotting library that helps in visualizing figures.

1.6 INTRODUCTION TO CSV FILES

A simple way to store big data sets is to use CSV files (comma separated files).

CSV files contain plain text and is a well-know format that can be read by everyone including Pandas. The csv file acts as the database where we store all the data related to the program for saving, accessing and modification of the data.

CHAPTER 2

OBJECTIVES

2.1 OBJECTIVE OF THE PROJECT

This project ‘ANALYSING IPL DATA’ aims to work with the data and visualize the data in the form of different charts and graphs using Python and its different modules. The software is designed in Python IDLE script mode and uses CSV file as background.

The objective of the project is to use Python DataFrame concept to find the team that won the most number of matches in a season. To find the missing data using NaN concept. To find the city that hosted the maximum number of IPL matches. To find the most winning team for each season and visualize it using bar charts. To find the on-field umpire with the maximum number of IPL matches. The data is stored in csv format and accessed using `read_csv()` function in our project to plot different graphs as per the choice of the end user.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 MINIMUM HARDWARE REQUIREMENTS

- x86 64-bit CPU (Intel / AMD architecture)
- 4 GB RAM
- 5 GB free disk space
- Display – Monochrome/VGA
- Standard QWERTY Keyboard
- Mouse

3.2 MINIMUM SOFTWARE REQUIREMENTS

- Modern Operating System:
 - Windows 8 or higher
 - Mac OS X 10.11 or higher, 64-bit
 - Linux: RHEL 6/7, 64-bit (almost all libraries also work in Ubuntu)
- Python IDLE v3.7 or later
- Spreadsheet program, such as Microsoft Excel, OpenOffice Calc for csv file operation.
- Modules – pandas, matplotlib.pyplot , numpy

CHAPTER 4

SYSTEM STUDY

4.1 EXISTING SYSTEM

The existing system forces us to rely on any software or online websites for storing the data and analysing the data for obtaining output. This could limit the possibilities as the software available for such data analysis is always restricted to the developer's choices and ideas.

Also, the data collected has to be entered in to such software, which is time consuming. It also leads to confusions, greater work load and at times is tedious too. Developing our own program that can be customised and changed as per our own decisions and requirements is not always possible with most existing software/system.

4.2 PROPOSED SYSTEM

In the proposed system, we are using Python to create data and store it with the help of a DataFrame and csv files. This can also be used for data visualization. The project is done in such a way that we incorporate functions to access the data as per the user's choice. This will help to obtain the desired output as per the choice of the user. We also provide the user to continue/exit the program at any point of time.

CHAPTER 5 SYSTEM DESIGN

5.1 SYSTEM DESIGN

System design is an approach to the creation of the system. It involves selection of system functions, drawing of the system, flowcharts, etc. It provides the understanding and procedural details necessary for implementing the system recommended in the feasibility studies.

The activities carried in the design phase are as follows:

1. Output design
2. Input design
3. Database design

5.2 OUTPUT DESIGN

Computer output is the means of directing information to the user. Efficiency in output design helps in decision making and improving the relationship between the system and the user. The design team defines the sketches into detailed descriptions of the output by planning the output with the specific medium. Printout is designed based on the output requirements of the user.

5.3 INPUT DESIGN

Input system is a phase system design. It is the process of converting the collected input data into a computer-based format. The aim of designing input data is to make data entry easy. This phase is often the first place for detecting and correcting errors, during data entry one should know the space allocated for each field, the field sequences that matches the source document and the format in which the data is to be entered. The objective of input data design is to create an input layout that is easy to follow and that which does not include operator errors.

5.4 DATABASE DESIGN

The user uses database to store integrated data that is input. It is organized so that various files can be accessed through a single reference. Here we use csv files to store the data and access the same using Python.

CHAPTER 6

SOURCE CODE

#Project name:Analysing IPL Data

```
import pandas as pd
```

```
import matplotlib.pyplot as pl
```

```
#read the csv file
```

```
ipl_matches_df = pd.read_csv('C://CLASS 12 IP PROJECT//matches.csv')
```

```
#input
```

```
def get_choice():
```

```
    try:
```

```
        return int(input("Enter your choice:\t"))
```

```
    except ValueError:
```

```
        print("Invalid input. Please enter a valid choice.")
```

```
    return None
```

```
#view data
```

```
def view_data():
```

```
    print("\n1.View DataFrame")
```

```
    print("2.View the DataFrame Info")
```

```
    print("3.View Statistical Data")
```

```
    print("4.Show Columns")
```

```
    c=get_choice()
```

```
    if c==1:
```

```
        print(ipl_matches_df)
```

```
    elif c==2:
```

```
        ipl_matches_df.info()
```

```
elif c==3:
    print(ipl_matches_df.describe())
elif c==4:
    print(ipl_matches_df.columns)
else:
    if c is not None:
        print("Invalid choice. Please enter a valid choice from 1 to 4.")
```

#check NaN

```
def check_null():
    print("\n1.First Row without NaN value")
    print("2.Null value count")
    c=get_choice()
    if c==1:
        #the first index that doesn't contain a NaN value
        print("\nThe first row without a NaN value is: ",end="")
        print(ipl_matches_df.umpire3.first_valid_index())
    elif c==2:
        #null values count
        print("\nColumn-wise NaN values count:\n")
        print(ipl_matches_df.isnull().sum())
    else:
        if c is not None:
            print("Invalid choice. Please enter a valid choice (1 or 2).")
```

```
#win per season

def win():

    print("\n1.Most wins in each season")

    print("2.Win per season")

    c=get_choice()

    if c==1:

        # team with the most wins in each season

        year = 2008

        teams_per_season =

            ipl_matches_df.groupby('season')['winner'].value_counts()

        most_wins_df = pd.DataFrame(columns=['year', 'team', 'wins'])

        print("\nWINS PER SEASON\n=====")

        for items in teams_per_season.items():

            if items[0][0]==year:

                #print(items)

                win_series = pd.DataFrame({

                    'year': [items[0][0]],

                    'team': [items[0][1]],

                    'wins': [items[1]],})

                most_wins_df = pd.concat([most_wins_df,win_series])

                year += 1

        print(most_wins_df)

    elif c==2:

        #win per season

        teams_per_season =

            ipl_matches_df.groupby('season')['winner'].value_counts()
```

```
        print(teams_per_season)
    else:
        if c is not None:
            print("Invalid choice. Please enter a valid choice(1 or 2).")

#visualization
def graph():
    print("\n1.Visualize win per season")
    print("2.Visualize venue hosted maximum games")
    print("3.Visualize most successful team")
    c=get_choice()
    if c==1:
        #create DataFrame for win per season
        year=2008

        teams_per_season=ipl_matches_df.groupby('season')['winner'].value_counts()

        win_per_season_df=pd.DataFrame(columns=['year', 'team','wins'])
        print("\nWINS PER SEASON\n=====")
        for items in teams_per_season.items():
            if items[0][0]==year:
                #print(items)
                win_series = pd.DataFrame({'year': [items[0][0]],\
                                           'team': [items[0][1]],\
                                           'wins': [items[1]]})
                win_per_season_df = pd.concat([win_per_season_df,win_series])
                year+=1
```

```
#visualization
pl.barh(win_per_season_df.team,win_per_season_df.wins,\
height=0.5,\
color=['r','g','b','k','y','c','m','g','b','k','y','c'])
pl.xlabel("Wins")
pl.ylabel("Teams")
pl.title("Season Winning")
pl.show()

elif c==2:
    #venue hosted max games
    venue_ser = ipl_matches_df['venue'].value_counts()
    venue_df = pd.DataFrame(columns=['venue', 'matches'])
    for items in venue_ser.items():
        temp_df = pd.DataFrame({
            'venue':[items[0]],
            'matches':[items[1]]
        })
        venue_df =pd.concat([venue_df,temp_df],ignore_index=True)
    #print(venue_df)

    #venue visualization
    c=list("rgbkycmyrgbkycmyrgbkycmyrgbkycmyrgbkycmyk")
    pl.bar(venue_df.venue,venue_df.matches,color=c)
    pl.xlabel("Stadium Name")
```

```
pl.ylabel("Number of matches")
pl.title("Hosted Matches by stadiums")
pl.xticks(rotation=90)
pl.grid(True)
pl.show()
elif c==3:
    #most successful teams
    team_wins_ser = ipl_matches_df['winner'].value_counts()
    team_wins_df = pd.DataFrame(columns=["team", "wins"])
    for items in team_wins_ser.items():
        temp_df1 = pd.DataFrame({
            'team':[items[0]],
            'wins':[items[1]]
        })
        team_wins_df = pd.concat([team_wins_df,temp_df1], ignore_index=True)
    #print(team_wins_df)

    #DV most successful teams
    c=list("rgbkycmcygrmkygb")
    pl.bar(team_wins_df.team,team_wins_df.wins,color=c)
    pl.xlabel("Team")
    pl.ylabel("Wins")
    pl.title("Most Successful Teams")
    pl.xticks(rotation=90)
    pl.grid(True)
```

```
pl.show()
else:
    if c is not None:
        print("Invalid choice. Please enter a valid choice from 1 to 3.")
#umpire details
def umpire():
    print("\n1.Umpire officiated most IPL matches")
    print("2.Top 5 Umpires")
    c=get_choice()
    if c==1:
        #umpire details - which umpire officiated the most IPL matches
        umpire1_ser = ipl_matches_df['umpire1'].value_counts()
        umpire2_ser = ipl_matches_df['umpire2'].value_counts()
        umpires_df = pd.concat([umpire1_ser, umpire2_ser], axis=1)
        umpire_ser = umpires_df.sum(axis=1)
        print(umpire_ser.sort_values(ascending=False).head(1).astype(int))

    elif c==2:
        #top 5 umpires officiated the matches
        umpire1_ser = ipl_matches_df['umpire1'].value_counts()
        umpire2_ser = ipl_matches_df['umpire2'].value_counts()
        umpires_df = pd.concat([umpire1_ser, umpire2_ser], axis=1)
        umpire_ser = umpires_df.sum(axis=1)
        umpire_df = pd.DataFrame(columns=["umpire", "matches"])
        for items in umpire_ser.items():
```

```
temp_df4 = pd.DataFrame({
    'umpire':[items[0]],
    'matches':[items[1]]
})

umpire_df= pd.concat([umpire_df,temp_df4], ignore_index=True)

print(umpire_df.sort_values('matches', ascending=False).head())

else:

    if c is not None:

        print("Invalid choice. Please enter a valid choice(1 or 2).")

#main menu
def main():

    while True:

        print("\nAnalysing IPL Data\n=====\n")

        print("Main Menu\n=====\n")

        print("1.View the Data")

        print("2.Check for NaN values")

        print("3.Win per season")

        print("4.Data Visualization")

        print("5.Umpire Details")

        print("6.Exit")

        ch=get_choice()

        if ch==1:

            view_data()

        elif ch==2:

            check_null()
```

```
elif ch==3:
    win()
elif ch==4:
    graph()
elif ch==5:
    umpire()
elif ch==6:
    print("\nThank you for using the program.")
    print("Exiting The Program...")
    exit()
else:
    if ch is not None:
        print("Invalid choice. Please enter a number from 1 to 6.")

main()
```

CHAPTER 7

OUTPUT WINDOW

7.1 MAIN MENU WINDOW

```
Analysing IPL Data
=====

Main Menu
=====

1.View the Data
2.Check for NaN values
3.Win per season
4.Data Visualization
5.Umpire Details
6.Exit
Enter your choice: |
```

7.2 CSV FILE

matches.csv - Excel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	id	season	city	date	team1	team2	toss_winnr	toss_decis	result	dl_applie	winner	win_by_r	win_by_w	player_of	venue	umpire1	umpire2	umpire3
1	1	2017	Hyderabad	05-04-2017	Sunrisers	Royal Cha	Royal Cha	field	normal	0	Sunrisers	35	0	Yuvraj Sin	Rajiv Ganc	AY Dandel	NJ Llong	
2	2	2017	Pune	06-04-2017	Mumbai Ir	Rising Pur	Rising Pur	field	normal	0	Rising Pur	0	7	SPD Smith	Maharash	A Nand Ki	S Ravi	
3	3	2017	Rajkot	07-04-2017	Gujarat Li	Kolkata Kr	Kolkata Kr	field	normal	0	Kolkata Kr	0	10	CA Lynn	Saurashtr	Nitin Men	CK Nandan	
4	4	2017	Indore	08-04-2017	Rising Pur	Kings XI Pi	Kings XI Pi	field	normal	0	Kings XI Pi	0	6	GJ Maxwe	Holkar Cri	AK Chaudl	C Shamshuddin	
5	5	2017	Bangalore	08-04-2017	Royal Cha	Delhi Dar	Royal Cha	bat	normal	0	Royal Cha	15	0	KM Jadhav	M Chinnaswamy	Stadium		
6	6	2017	Hyderabad	09-04-2017	Gujarat Li	Sunrisers	Sunrisers	field	normal	0	Sunrisers	0	9	Rashid Kh	Rajiv Ganc	A Deshm	NJ Llong	
7	7	2017	Mumbai	09-04-2017	Kolkata Kr	Mumbai Ir	Mumbai Ir	field	normal	0	Mumbai Ir	0	4	N Rana	Wankhed	Nitin Men	CK Nandan	
8	8	2017	Indore	10-04-2017	Royal Cha	Kings XI Pi	Royal Cha	bat	normal	0	Kings XI Pi	0	8	AR Patel	Holkar Cri	AK Chaudl	C Shamshuddin	
9	9	2017	Pune	11-04-2017	Delhi Dar	Rising Pur	Rising Pur	field	normal	0	Delhi Dar	97	0	SV Samsot	Maharash	AY Dandel	S Ravi	
10	10	2017	Mumbai	12-04-2017	Sunrisers	Mumbai Ir	Mumbai Ir	field	normal	0	Mumbai Ir	0	4	JJ Bumrah	Wankhed	Nitin Men	CK Nandan	
11	11	2017	Kolkata	13-04-2017	Kings XI Pi	Kolkata Kr	Kolkata Kr	field	normal	0	Kolkata Kr	0	8	SP Narine	Eden Gard	A Deshm	NJ Llong	
12	12	2017	Bangalore	14-04-2017	Royal Cha	Mumbai Ir	Mumbai Ir	field	normal	0	Mumbai Ir	0	4	KA Pollarc	M Chinnas	KN Anantl	AK Chaudhary	
13	13	2017	Rajkot	14-04-2017	Rising Pur	Gujarat Li	Gujarat Li	field	normal	0	Gujarat Li	0	7	AJ Tye	Saurashtr	A Nand Ki	S Ravi	
14	14	2017	Kolkata	15-04-2017	Kolkata Kr	Sunrisers	Sunrisers	field	normal	0	Kolkata Kr	17	0	RV Uthap	Eden Gard	AY Dandel	NJ Llong	
15	15	2017	Delhi	15-04-2017	Delhi Dar	Kings XI Pi	Delhi Dar	bat	normal	0	Delhi Dar	51	0	CJ Anders	Feroz Shal	YC Barde	Nitin Menon	
16	16	2017	Mumbai	16-04-2017	Gujarat Li	Mumbai Ir	Mumbai Ir	field	normal	0	Mumbai Ir	0	6	N Rana	Wankhed	A Nand Ki	S Ravi	
17	17	2017	Bangalore	16-04-2017	Rising Pur	Royal Cha	Royal Cha	field	normal	0	Rising Pur	27	0	BA Stokes	M Chinnas	KN Anantl	C Shamshuddin	
18	18	2017	Delhi	17-04-2017	Delhi Dar	Kolkata Kr	Delhi Dar	bat	normal	0	Kolkata Kr	0	4	NM Coultr	Feroz Shal	Nitin Men	CK Nandan	
19	19	2017	Hyderabad	17-04-2017	Sunrisers	Kings XI Pi	Kings XI Pi	field	normal	0	Sunrisers	5	0	B Kumar	Rajiv Ganc	AY Dandel	A Deshmukh	
20	20	2017	Rajkot	18-04-2017	Royal Cha	Gujarat Li	Gujarat Li	field	normal	0	Royal Cha	21	0	CH Gayle	Saurashtr	S Ravi	VK Sharma	
21	21	2017	Hyderabad	19-04-2017	Sunrisers	Delhi Dar	Sunrisers	bat	normal	0	Sunrisers	15	0	KS Willian	Rajiv Ganc	CB Gaffan	NJ Llong	
22	22	2017	Indore	20-04-2017	Kings XI Pi	Mumbai Ir	Mumbai Ir	field	normal	0	Mumbai Ir	0	8	JC Buttler	Holkar Cri	M Erasmu	C Shamshuddin	

7.3 DATAFRAME DISPLAYED FROM CSV

```

1.View DataFrame
2.View the DataFrame Info
3.View Statistical Data
4.Show Columns

Enter your choice:      1

   id  season  ...      umpire2      umpire3
0    1   2017  ...      NJ Llong      NaN
1    2   2017  ...      S Ravi      NaN
2    3   2017  ...      CK Nandan      NaN
3    4   2017  ...  C Shamshuddin      NaN
4    5   2017  ...      NaN      NaN
..   ...    ...  ...      ...      ...
751 11347   2019  ...      O Nandan      S Ravi
752 11412   2019  ...      Nitin Menon      Ian Gould
753 11413   2019  ...      NaN      NaN
754 11414   2019  ...  Bruce Oxenford  Chettithody Shamshuddin
755 11415   2019  ...      Ian Gould      Nigel Llong

```

7.4 RUN-TIME SAMPLE OUTPUT WINDOWS

```

Enter your choice:      2

Column-wise NaN values count:

id           0
season       0
city         7
date         0
team1        0
team2        0
toss_winner  0
toss_decision 0
result       0
dl_applied   0
winner       4
win_by_runs  0
win_by_wickets 0
player_of_match 4
venue        0
umpire1      2

```

WINS PER SEASON

=====

	year	team	wins
0	2008	Rajasthan Royals	13
0	2009	Delhi Daredevils	10
0	2010	Mumbai Indians	11
0	2011	Chennai Super Kings	11
0	2012	Kolkata Knight Riders	12
0	2013	Mumbai Indians	13
0	2014	Kings XI Punjab	12
0	2015	Chennai Super Kings	10
0	2016	Sunrisers Hyderabad	11
0	2017	Mumbai Indians	12
0	2018	Chennai Super Kings	11
0	2019	Mumbai Indians	11

4.Data Visualization

5.Umpire Details

6.Exit

Enter your choice: 5

1.Umpire officiated most IPL matches

2.Top 5 Umpires

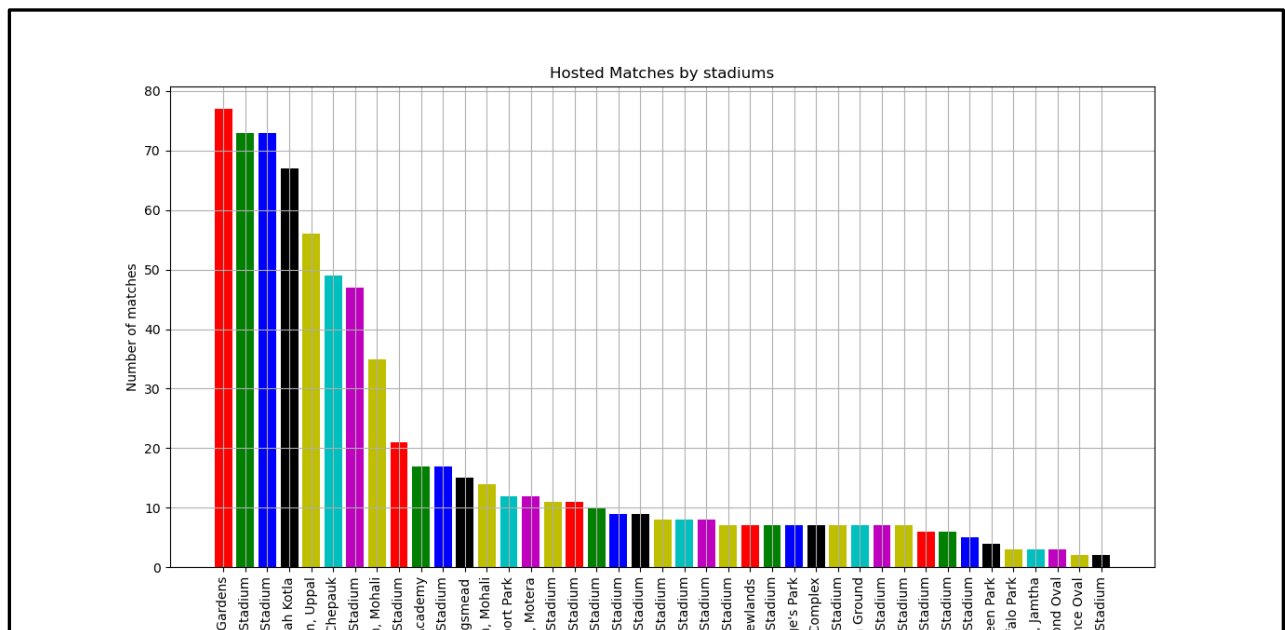
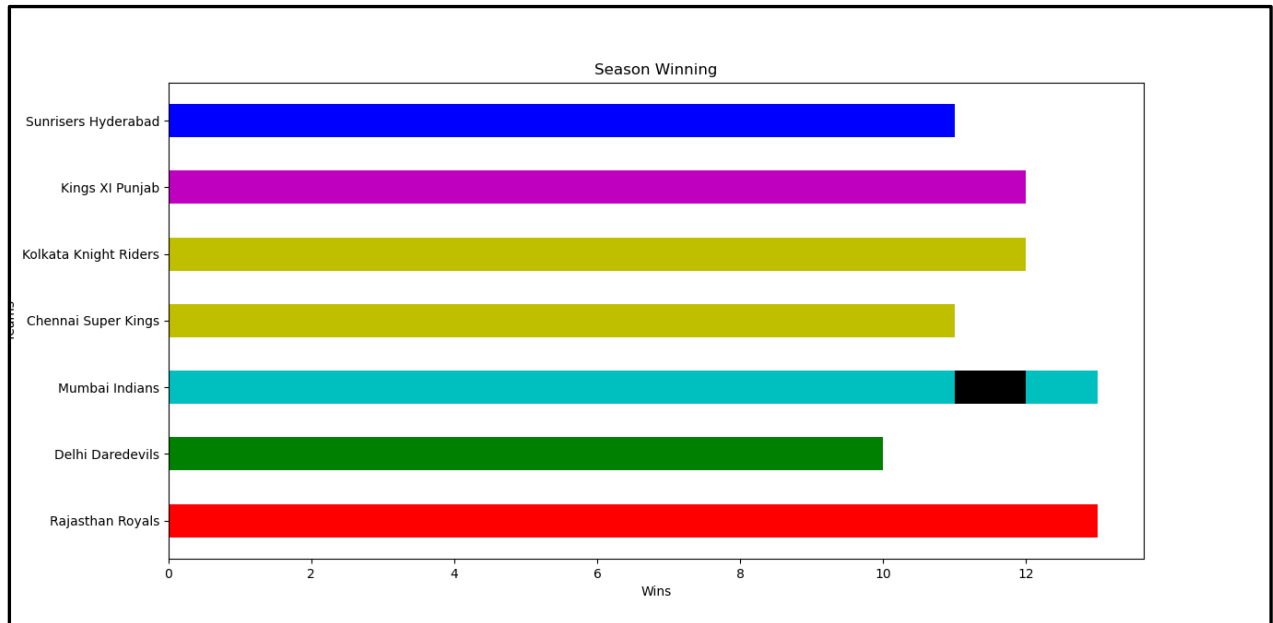
Enter your choice: 2

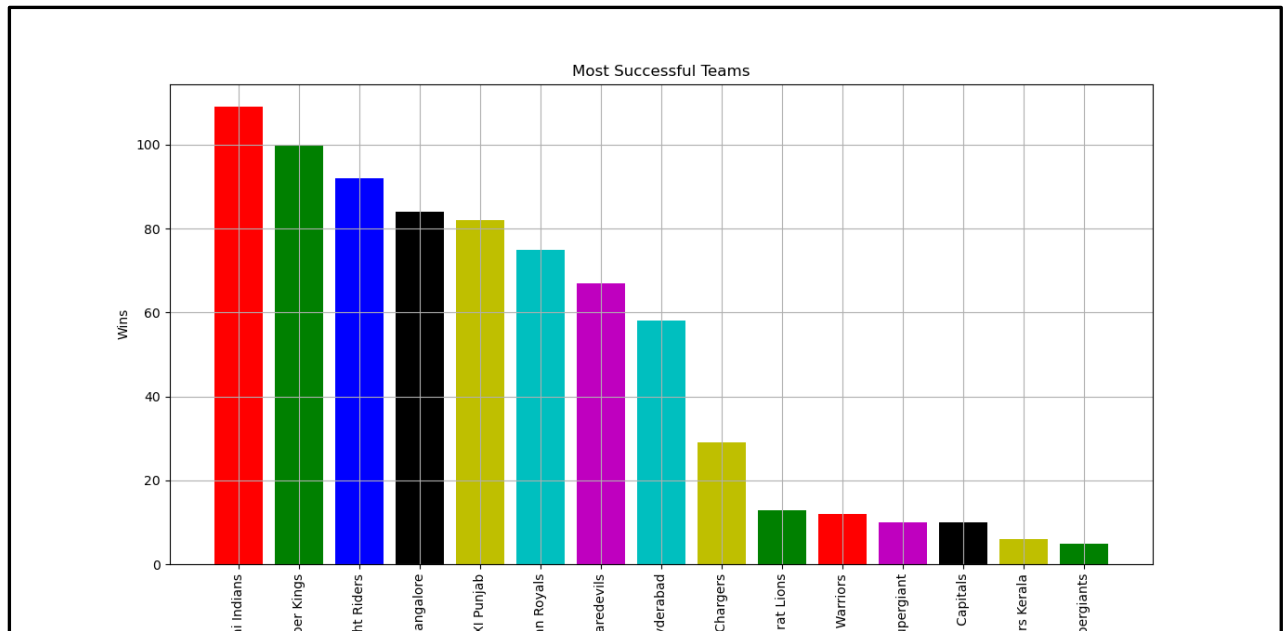
	umpire	matches
2	S Ravi	106.0
0	HDPK Dharmasena	87.0
11	C Shamshuddin	73.0
3	AK Chaudhary	58.0
56	SJA Taufel	55.0

Analysing IPL Data

=====

7.5 OUTPUT CHART





CHAPTER 8

CONCLUSION

Data Science / Analytics is all about finding valuable insights from the given dataset. In short, finding answers that could help business. This program is used to analyse the data store in a csv file using Python's DataFrame and visualization techniques by matplotlib, specifically related to IPL data. The program generates bar graph based on the choice given by user. Using the method, the data related to the I.P.L can be processed to obtain the details like win per season by each team, Umpire details, visualizing the matches played in each venue, finding out number of matches won by each team season-wise and also the wins per team.

The program can be used in general analysis and determining the success rate by each team. It also tells venue factor in winning the matches in each season. That tells us two things: Python Expressions are very crispy in terms of its syntax and the second thing is, it does not need tens or hundreds of lines to bring out a valuable insight — just a few lines could do the magic — if you're asking the right question!

CHAPTER 9 REFERENCES

1. Informatics Practices - Class XII By : Sumita Arora
2. Website: <https://www.w3resource.com>
3. Website : https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.read_csv.html
4. Website : www.python.org